

Project Close-out Report

Name of Project and Project URL

Pre-IPO Perpetuals — Trade Private Companies on Cardano

Project URL: <https://projectcatalyst.io/funds/14/cardano-use-cases-concepts/pre-ipo-perpetuals-trade-private-companies-on-cardano>

Project Number

Project ID: 1400062

Name of Project Manager

Dinesh Kumar

Date Project Started

October 21, 2025

Date Project Completed

May 14, 2026

Challenge KPIs and How the Project Addressed Them

The project delivered a Cardano-native marketplace for **synthetic valuation-index exposure** to Anthropic, OpenAI, and SpaceX, backed by an Aiken Plutus V3 validator and an off-chain Lucid Evolution service. All sources, the compiled blueprint (`plutus.json`), documentation, and Preprod evidence are committed to the public repository: github.com/dextrlabs-dev/Pre-IPO-Perpetuals-Cardano.

Open-source infrastructure — single Aiken validator (`preipo`) covering `Oracle` , `Mint` , `Redeem` , and `Swap` redeemers; Deno HTTP API with OpenAPI / Swagger; in-process oracle worker; Vite + React trader UI; reproducible Preprod evidence pipeline.

Cardano-native DeFi surface — one UTxO per market at a shared validator address, with strict transition rules (preserved native value, monotonic `seq` , bounded oracle moves via `max_variance_bps = 2000` , TWAV blending `9/10` , pool deviation guard).

End-to-end Preprod validation — full `lock → oracle → mint → swap → redeem` cycle executed across three markets, with all 15 transactions landing on-chain successfully.

Project KPIs and How the Project Addressed Them

Milestone	Deliverables	Status
M1 Technical Architecture & Initiation	Technical Architecture PDF + Project Initiation PDF	Delivered (Dec 2025)
M2 MVP Development	MVP technical documentation, OpenAPI spec, Aiken validator, off-chain service, frontend, demo video	Delivered (Jan 2026)
M3 Testing, Launch & Close-Off	Final Technical Documentation, Post-Launch Operations, Close-Out Report (this doc), tagged <code>final-v1.0</code>	Delivered (May 2026)

Key Achievements

- One of the first end-to-end Cardano-native marketplaces for **synthetic valuation-index exposure to private companies**, with all economic state encoded in inline datums and all transitions guarded by a single Aiken validator.
- 15 / 15 Preprod transactions** submitted and confirmed across all three markets and all four redeemer paths — full evidence in [docs/preprod-run-2026-03-23T10-48-17-980Z.csv](#) .
- Three independent markets (Anthropic, OpenAI, SpaceX) running concurrently on the same validator address with full economic separation per UTxO.
- Complete operational profile documented: launch procedures, SLOs, incident-response playbooks, governance — in [docs/POST-LAUNCH-OPERATIONS.md](#) .
- Public release of the full stack (Aiken validator, Deno API, React UI, CLI runners, evidence pipeline) under the repository's open-source posture.

Impact

- All three funded milestones completed and approved.
- Functional MVP deployment exercised on Cardano Preprod with public, link-verifiable transaction evidence.
- Reference Aiken Plutus V3 validator demonstrating composable **single-validator multi-market** design.
- Lucid Evolution-based transaction service reusable by other Cardano DeFi builders.
- Documentation set covering architecture, MVP, launch, operations, and close-off — all version-tagged.

Why is this Project Important?

This project demonstrates that Cardano can host **complex, multi-market economic systems** under a single Aiken validator with bounded-move oracle updates, constant-product swaps, and collateral accounting — without secondary chains, bridges, or off-chain settlement layers. As Cardano DeFi grows, there is increasing demand for composable single-validator designs, patterns for routing external reference data into bounded on-chain updates, and open reference implementations that combine Aiken, Lucid Evolution, and modern frontend tooling end-to-end. The Pre-IPO Perpetuals marketplace addresses these through its single-validator multi-market design, its TWAV + variance-bounded oracle, and its full open-source release including documentation, CLI, and a Preprod evidence pipeline.

Lessons Learned & Next Steps

Lessons learned — single-validator multi-market is a strong pattern (small validator, linear operational scale); datum-authoritative accounting lets USD-style display numbers live entirely off-chain without compromising on-chain correctness; bounded oracle + TWAV absorbed feed jitter across all three markets in the Preprod batch; versioning evidence inside the repository keeps the trader UI honest.

Next steps (out of current scope, documented in Post-Launch Operations §4.3) — multi-trader markets via script-credential `PerpDatum.trader`; distributed oracle keyset (multisig redeemer); native USDCx asset to retire the off-chain faucet ledger; governance-controlled reference datum for validator parameters.

Links to Other Relevant Project Sources or Documents

Resource	Link
Project Repository	github.com/dextrlabs-dev/Pre-IPO-Perpetuals-Cardano
Technical Architecture (M1)	Technical Documentation Pre-IPO Marketplace.pdf
Project Initiation (M1)	Initiation Document Pre-IPO Marketplace.pdf
MVP Technical Documentation (M2)	docs/MVP-TECHNICAL-DOCUMENTATION.md
Final Technical Documentation (M3)	docs/FINAL-TECHNICAL-DOCUMENTATION.md
Post-Launch Operations (M3)	docs/POST-LAUNCH-OPERATIONS.md
Preprod Evidence CSV	docs/preprod-run-2026-03-23T10-48-17-980Z.csv
Validator Source	validators/preipo.ak
Release Tag	final-v1.0

Project Close-out Video

[Close-out video URL – to be added once recorded]